

phase2-week4-day2-retrieval-api

Vector Search:

The screenshot shows a REST client interface with a workspace named "My Workspace". The active collection is "RAG MongoDB - Complete Ingestion & Retrieval". The selected endpoint is "09. Vector Search" with a POST method. The URL is "({{baseUrl}}) /api/search". The request body is a JSON object:

```
1 {
2   "query": "{{searchQuery}}",
3   "limit": {{searchLimit}},
4   "filters": {
5     "module": "",
6     "priority": "",
7     "risk": "",
8     "automationManual": ""
9   }
}
```

The response is a 200 OK status with a response time of 4.98 s and a size of 9.01 KB. The response body is a JSON object:

```
32 {
33   "_id": "6a378acd65672f302484dffc",
34   "module": "AHC",
35   "id": "TC_2617",
36   "preRequisites": "No Pre-requisite",
37   "title": "MasterSetup->Map Login Counter ->Delete",
38   "description": "To verify the user able to delete Map Login Counter.",
39   "steps": "\r\n1.Launch the application\r\n2.Login to the application with valid credentials.\r\n3.Click on Patient Service Tab and select AHC from drop down.\r\n4.Navigate to Functions -> MasterSetup\r\n5.Map Login Counter \r\n6.Click"
}
```

BM25 Search:

The screenshot shows a REST client interface with a workspace named "My Workspace". The active collection is "RAG MongoDB - Complete Ingestion & Retrieval". The selected endpoint is "10. BM25 Full-Text Search" with a POST method. The URL is "({{baseUrl}}) /api/search/bm25". The request body is a JSON object:

```
1 {
2   "query": "{{searchQuery}}",
3   "limit": {{searchLimit}},
4   "filters": {
5     "module": "",
6     "priority": "",
7     "risk": ""
8   }
9 }
```

The response is a 200 OK status with a response time of 2.74 s and a size of 8.06 KB. The response body is a JSON object:

```
12 {
13   "_id": "6a3789f965672f302484ccb9",
14   "module": "Admin",
15   "id": "TC_2001",
16   "title": "Functionality",
17   "description": "To verify the user able to Create a Functionality to the Business module.",
18   "steps": "1.Launch the application\r\n2.Login to the application with valid credentials.\r\n3.Click on External Service Tab and select Admin from the drop down.\r\n4.Click on Navigation Menu -> Masters ->Functionality.\r\n5.Click on the Functionality.\r\n6.Enter Functionality Name, Business Module and FunctionalityAlias Name.\r\n7.Click on \"SAVE\" button."
}
```

Hybrid Search:

The screenshot shows the RAG MongoDB interface with the following details:

- Left Sidebar:** Collections list including GenAI-May, GenAI-May-Home Work, Jenkins_Requests, JIRA, jiraTest Copy, Langflow_Workflows, RAG MongoDB - Complete Ingestion & Retrieval (with sub-items INGESTION and RETRIEVAL), POST 09. Vector Search, POST 10. BM25 Full-Text Search, **POST 11. Hybrid Search (BM25 + Vector)**, POST 12. Rerank Search (Groq AI LLM Reranking), POST 13. Preprocess Query, POST 14. Analyze Query, POST-RETRIEVAL, UTILITIES, GET 00. Health Check, ENVIRONMENTS, SPECS, and FLOWS.
- Main Panel:** A POST request to `{{baseUrl}}/api/search/hybrid` is shown. The request body is a JSON object with fields: `query`, `limit`, `bm25Weight`, `vectorWeight`, and `filters`. The response status is **200 OK** with a response time of 5.88 s and a size of 9.52 KB. The response body is a JSON array with one object containing fields: `_id`, `module`, `id`, `title`, `description`, and `steps`.
- Right Panel:** A 'New Chat' section with a text input field and a 'Send' button. Below it, a 'Compare search performance' section with a 'Test Hybrid Search' button. At the bottom, a 'Describe what you need. Press @ for context, / for Skills.' prompt is visible.

Rerank Search:

The screenshot shows the RAG MongoDB interface with the following details:

- Left Sidebar:** Collections list including GenAI-May, GenAI-May-Home Work, Jenkins_Requests, JIRA, jiraTest Copy, Langflow_Workflows, RAG MongoDB - Complete Ingestion & Retrieval (with sub-items INGESTION and RETRIEVAL), POST 09. Vector Search, POST 10. BM25 Full-Text Search, POST 11. Hybrid Search (BM25 + Vector), **POST 12. Rerank Search (Groq AI LLM Reranking)**, POST 13. Preprocess Query, POST 14. Analyze Query, POST-RETRIEVAL, UTILITIES, GET 00. Health Check, ENVIRONMENTS, SPECS, and FLOWS.
- Main Panel:** A POST request to `{{baseUrl}}/api/search/rerank` is shown. The response status is **200 OK** with a response time of 8.46 s and a size of 7.55 KB. The response body is a JSON array with one object containing fields: `_id`, `module`, `id`, `title`, `description`, `steps`, `expectedResults`, `priority`, `risk`, and `bm25Score`.
- Right Panel:** A 'New Chat' section with a text input field and a 'Send' button. Below it, a 'Compare search performance' section with a 'Test Hybrid Search' button. At the bottom, a 'Describe what you need. Press @ for context, / for Skills.' prompt is visible.

My Workspace

Search

POST Gene POST Chat POST 09. V POST 10. B POST 11. H POST 12. R + No environment

RAG MongoDB - Complete Ingestion & Retrieval

POST 09. Vector Search

POST 10. BM25 Full-Text Search

POST 11. Hybrid Search (BM25 + Vector)

POST 12. Rerank Search (Groq AI LLM Reranking)

POST 13. Preprocess Query

POST 14. Analyze Query

POST-RETRIEVAL

UTILITIES

GET 00. Health Check

ENVIRONMENTS

SPECS

FLOWS

POST

{{baseUri}} /api/search/rerank

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

raw JSON Schema Beautify

Body

200 OK 8.46 s 7.55 KB Save Response

```
{
  "module": "Radiology",
  "id": "TC_3702",
  "title": "Radiology-Masters-Login Modality Master-Delete",
  "description": "Verify user is able to Delete Login Modality Master",
  "steps": "1.Launch the Application.\r\n2.Login to the application with valid credentials\r\n3.Click on Patient Service Tab and select Radiology from the drop down.\r\n4.Navigate to Module ->Diagnostics and Imaging ->Masters ->Login Modality Master\r\n5. Select Modality\r\n6. Enter Employee ID\r\n7. Click on View and Click on Delete Icon\r\n8. Click ok on the popup displayed",
  "expectedResults": "Record Deleted Message should be displayed",
  "priority": "P1",
  "risk": "Critical",
  "vectorScore": 0.8802387118339539,
  "rerankScore": 0.88,
  "originalRank": 59
}
```

Compare search performance

Test Hybrid Search

Describe what you need. Press @ for context, / for Skills.

Auto

Connect Git Console Terminal Import Complete

Globals Vault Tools

Preprocess Query:

My Workspace

Search

POST Chat POST 09. V POST 10. B POST 11. H POST 12. R POST 13. P + No environment

RAG MongoDB - Complete Ingestion & Retrieval

POST 09. Vector Search

POST 10. BM25 Full-Text Search

POST 11. Hybrid Search (BM25 + Vector)

POST 12. Rerank Search (Groq AI LLM Reranking)

POST 13. Preprocess Query

POST 14. Analyze Query

POST-RETRIEVAL

UTILITIES

GET 00. Health Check

ENVIRONMENTS

SPECS

FLOWS

POST

{{baseUri}} /api/search/preprocess

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

raw JSON Schema Beautify

Body

200 OK 6.59 s 1019 B Save Response

```
{
  "query": "{{searchQuery}}",
  "options": {
    "enableAbbreviations": true,
    "enableSynonyms": true,
    "maxSynonymVariations": 5,
    "smartExpansion": false,
    "preserveTestCaseIds": true
  }
}
```

Body

```
{
  "original": "login functionality test cases",
  "normalized": "login functionality test cases",
  "abbreviationExpanded": "login functionality test cases",
  "synonymExpanded": [
    "login functionality test cases",
    "signin functionality test cases",
    "authenticate functionality test cases",
    "access functionality test cases",
    "logon functionality test cases"
  ]
}
```

Add tests to Preprocess Query

Run retrieval pipeline tests

Describe what you need. Press @ for context, / for Skills.

Auto

Connect Git Console Terminal Import Complete

Globals Vault Tools

My Workspace

Search

POST Chat POST 09. V: POST 10. B: POST 11. H: POST 12. R: POST 13. P: +

POST 13. Preprocess Query

POST {{baseUrl}} /api/search/preprocess

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

raw JSON Schema Beautify

Body

```
1 {
2   "term": "login",
3   "synonyms": [
4     "signin",
5     "authenticate",
6     "access",
7     "login"
8   ],
9   "position": 0
10 }
11
```

200 OK - 6.59 s - 1019 B

Save Response

JSON Preview Visualize

```
15 {
16   "term": "login",
17   "synonyms": [
18     "signin",
19     "authenticate",
20     "access",
21     "login"
22   ],
23   "position": 0
24 },
25 {
26   "term": "test",
27   "synonyms": [
28     "examination",
29     "investigation",
30     "analysis",
31     "screening"
32   ],
33   "position": 0
34 }
35
```

Add tests to Preprocess Query

Run retrieval pipeline tests

Describe what you need. Press @ for context, / for Skills.

Auto

Connect Git Console Terminal Import Complete

Globals Vault Tools

Analyze Query:

My Workspace

Search

POST 09. V: POST 10. B: POST 11. H: POST 12. R: POST 13. P: POST 14. A: +

POST 14. Analyze Query

POST {{baseUrl}} /api/search/analyze

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

raw JSON Schema Beautify

Body

```
1 {
2   "query": "{{searchQuery}}"
3 }
```

200 OK - 11 ms - 781 B

Save Response

JSON Preview Visualize

```
1 {
2   "original": "login functionality test cases",
3   "normalized": "login functionality test cases",
4   "analysis": {
5     "hasTestCaseIds": false,
6     "testCaseIds": [],
7     "hasAbbreviations": false,
8     "abbreviations": [],
9     "hasSynonymOpportunities": true,
10    "synonymOpportunities": [
11      {
12        "term": "login",
13        "synonyms": [
14          "signin",
15          "authenticate",
16          "access"
17        ]
18      }
19    ]
20  }
21 }
```

Add tests to Preprocess Query

Run retrieval pipeline tests

Describe what you need. Press @ for context, / for Skills.

Auto

Connect Git Console Terminal Import Complete

Globals Vault Tools

My Workspace

COLLECTIONS

GenAI-May

GenAI-May-Home Work

Jenkins_Requests

JIRA

jiraTest Copy

Langflow_Workflows

RAG MongoDB - Complete Ingestion & Retrieval

INGESTION

RETRIEVAL

POST 09. Vector Search

POST 10. BM25 Full-Text Search

POST 11. Hybrid Search (BM25 + Vector)

POST 12. Rerank Search (Groq AI LLM Reranking)

POST 13. Preprocess Query

POST 14. Analyze Query

POST-RETRIEVAL

UTILITIES

GET 00. Health Check

ENVIRONMENTS

SPECS

FLOWES

Search

Invite Upgrade

No environment

AI

New Chat

RAG MongoDB - Complete Ingestion & Retr...

14. Analyze Query

Save Share

POST

{{baseUrl}} /api/search/analyze

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

Body

200 OK 11 ms 781 B Save Response

JSON Preview Visualize

```
9      "hasSynonymOpportunities": true,
10
11      "synonymOpportunities": [
12        {
13          "term": "login",
14          "synonyms": [
15            "signin",
16            "authenticate",
17            "access",
18            "logon"
19          ],
20          "position": 0
21        },
22        {
23          "term": "test",
24          "synonyms": [
25            "examination",
26            "investigation",
27            "analysis",
28            "screening"
29          ],
30          "position": 2
31        }
32      ]
33    }
34  ]
35}
```

Add tests to Preprocess Query

Run retrieval pipeline tests

Describe what you need. Press @ for context, / for Skills.

Auto

Connect Git Console Terminal 0 0 Import Complete

Globals Vault Tools